



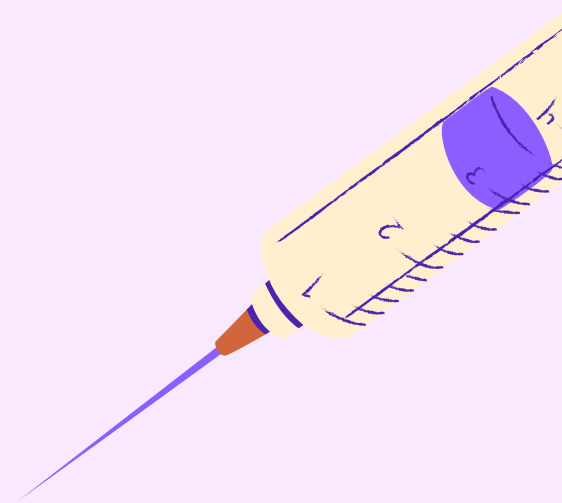
HEALTHCARE PRESENTATION

Presented by Team 4

INTRODUCTION

Healthcare providers routinely make treatment decisions without a reliable way to anticipate how a patient will respond — whether treatment will fail, lead to partial improvement, or result in full recovery. This uncertainty affects everything from resource allocation to patient counseling and follow-up care planning.

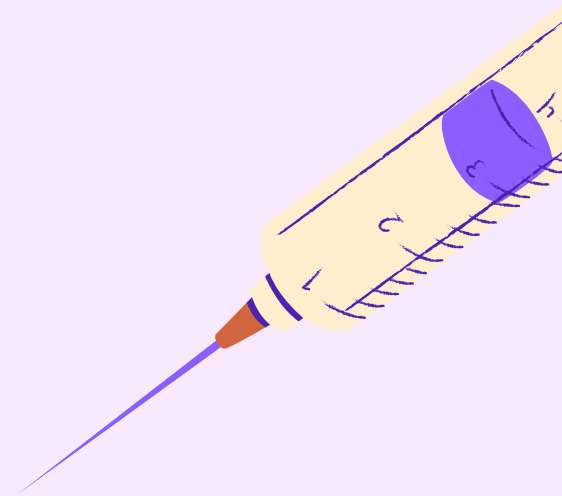
This project applies supervised machine learning to a dataset of 5,000 patient records, combining clinical vitals (blood pressure, heart rate, temperature), laboratory results (cholesterol, blood sugar, liver and kidney function), and behavioral/demographic factors (age, BMI, smoking status, exercise habits, treatment adherence) to predict treatment outcome — classified as failure, improvement, or recovered.

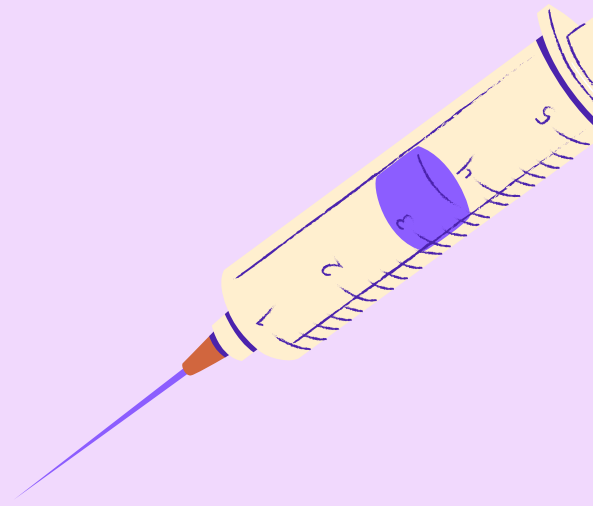


INTRODUCTION

The objective is twofold: first, to build a classification model capable of identifying patients at risk of treatment failure early, enabling more targeted clinical intervention; and second, to uncover which clinical and behavioral factors most strongly influence recovery, offering interpretable insights alongside predictive performance.

We evaluate multiple modeling approaches — including Logistic Regression, K-Nearest Neighbors, and XGBoost — with particular attention to the dataset's significant class imbalance (only 8% of patients fully recover), using macro-averaged F1 score rather than raw accuracy to ensure the model is genuinely useful for identifying the minority, clinically critical outcomes rather than simply favoring the majority class.





PREPROCESSING

In the preprocessing phase, we handled missing values across numerical features like blood pressure and BMI using the median to stay robust against outliers. For categorical columns like medication and alcohol intake, we filled them using mode and categorical defaults. We also derived missing treatment start dates mathematically.



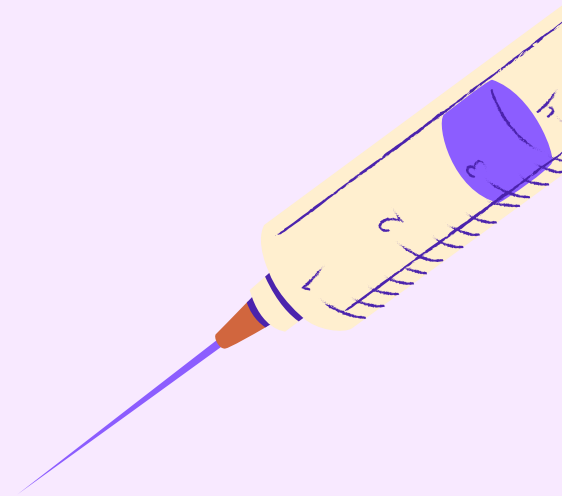
EDA

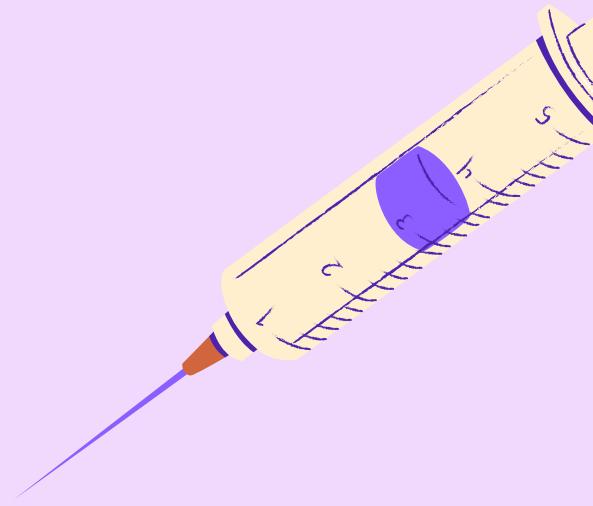
A

During our exploratory data analysis, we started with univariate analysis to inspect patient demographics, finding a uniform distribution across age groups and genders.

B

Then, in bivariate and multivariate analysis, we explored how clinical metrics like blood sugar correlate with treatment outcomes and diagnoses, and checked our correlation heatmap to ensure no high collinearity between variables.





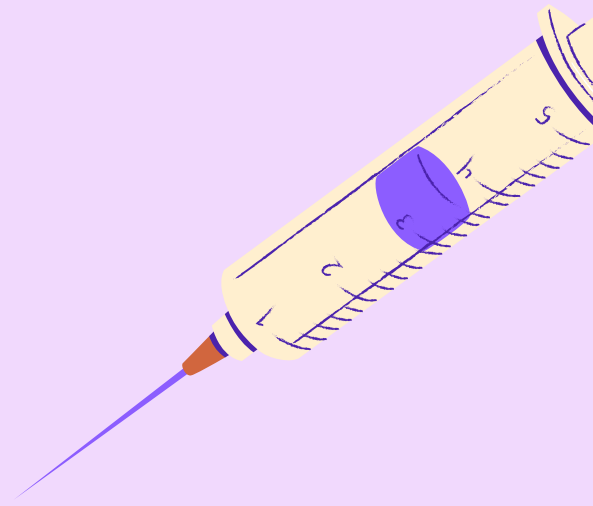
SPLITTING DATA

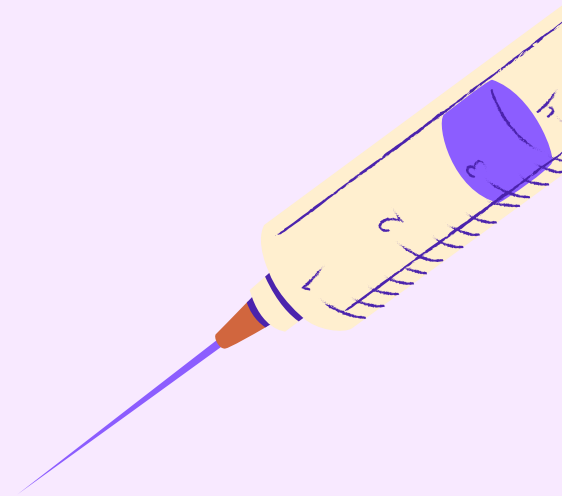
To evaluate model performance fairly, the dataset was split into training (80%) and testing (20%) sets using a stratified split, ensuring that the proportion of each treatment outcome — failure, improvement, and recovered — remained consistent across both sets. This was essential given the dataset's class imbalance (67% failure, 25% improvement, only 8% recovered); without stratification, the smaller "recovered" class risked being underrepresented or even absent from the test set, which would distort evaluation results.



MODEL 1: LOGISTIC REGRESSION

Logistic Regression was used as the baseline model due to its simplicity and interpretability. It estimates the probability of each treatment outcome class using a linear combination of the input features. Class weighting (`class_weight='balanced'`) was applied to counteract the dataset's imbalance, giving the minority "recovered" class proportionally more influence during training. This model provides a clear reference point for evaluating whether more complex models offer meaningful improvement.





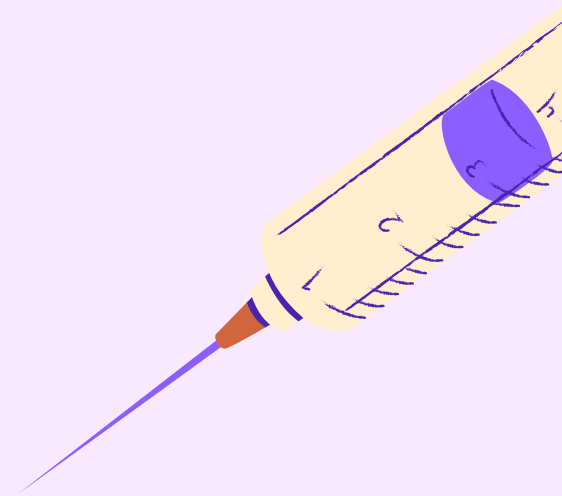
CONFUSION MATRIX

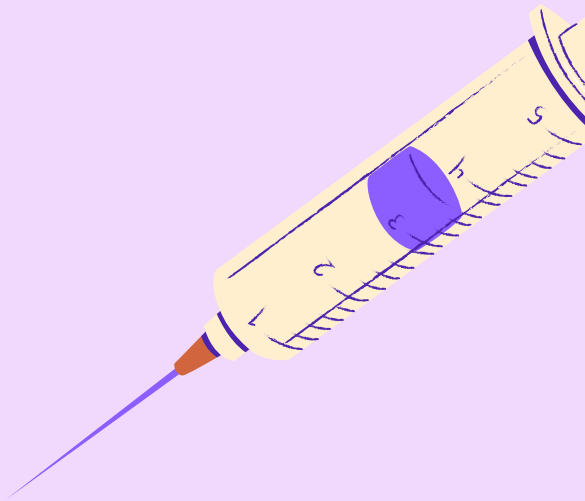
	PRECISION	RECALL	F1-SCORE	SUPPORT
ACCURACY	0.94			1000
MACRO AVG	0.87	0.92	0.89	1000
WEIGHTED AVG	0.95	0.94	0.94	1000



MODEL 2: K-NEAREST NEIGHBORS (KNN)

KNN classifies each patient based on the treatment outcomes of the "k" most similar patients in the training data, measured by distance across all clinical and behavioral features. Because KNN relies on distance calculations, feature scaling (standardization) was applied beforehand to prevent variables with larger numeric ranges — such as blood sugar or hospital stay duration — from dominating the distance metric. This model captures non-linear patterns that Logistic Regression may miss, though it is more sensitive to noise and class imbalance.

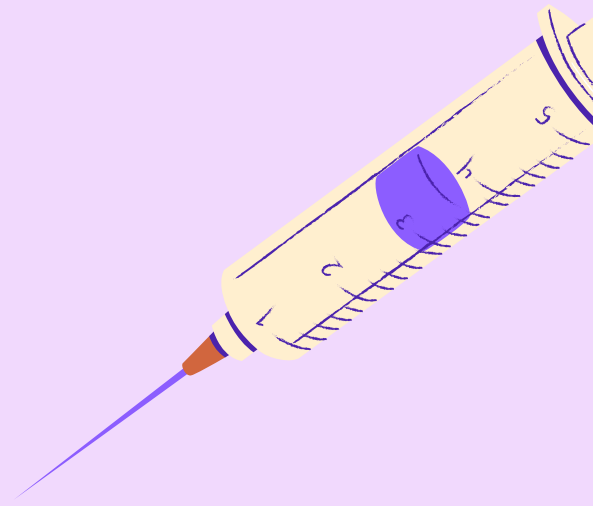




CONFUSION MATRIX

	PRECISION	RECALL	F1-SCORE	SUPPORT
ACCURACY			0.92	1000
MACRO AVG	0.89	0.78	0.80	1000
WEIGHTED AVG	0.92	0.92	0.91	1000

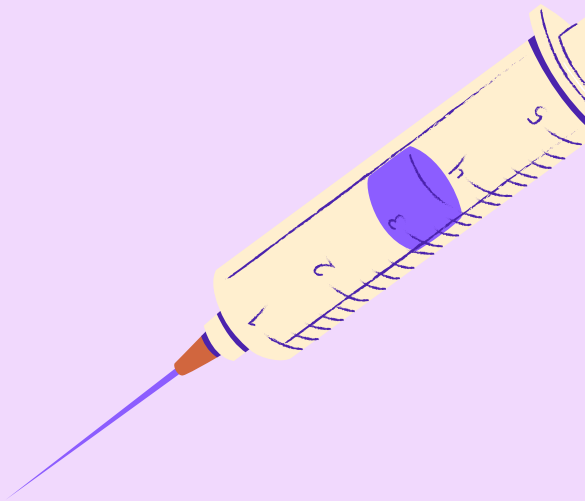




MODEL 3: XGBOOST

XGBoost (Extreme Gradient Boosting) is an ensemble model that builds a sequence of decision trees, where each new tree corrects the errors of the previous ones. It was selected for its strong performance on structured/tabular healthcare data and its ability to model complex, non-linear relationships between features without requiring feature scaling. Sample weighting was applied during training to address class imbalance, ensuring the model didn't disproportionately favor the majority "failure" class. XGBoost also offers feature importance scores, providing interpretability into which clinical factors most influence treatment outcomes.





CONFUSION MATRIX

	PRECISION	RECALL	F1-SCORE	SUPPORT
ACCURACY			1.00	1000
MACRO AVG	1.00	1.00	1.00	1000
WEIGHTED AVG	1.00	1.00	1.00	1000



MODEL 4 : SVM

Tries to draw a boundary (line or curve) between classes that stays as far as possible from every data point.

Works well on balanced data, but here it completely failed on the small "recovered" class (0% recall) because that class is rare. Result: 86%.

SVM (Support Vector Machine)

Multi-class classification of treatment outcome

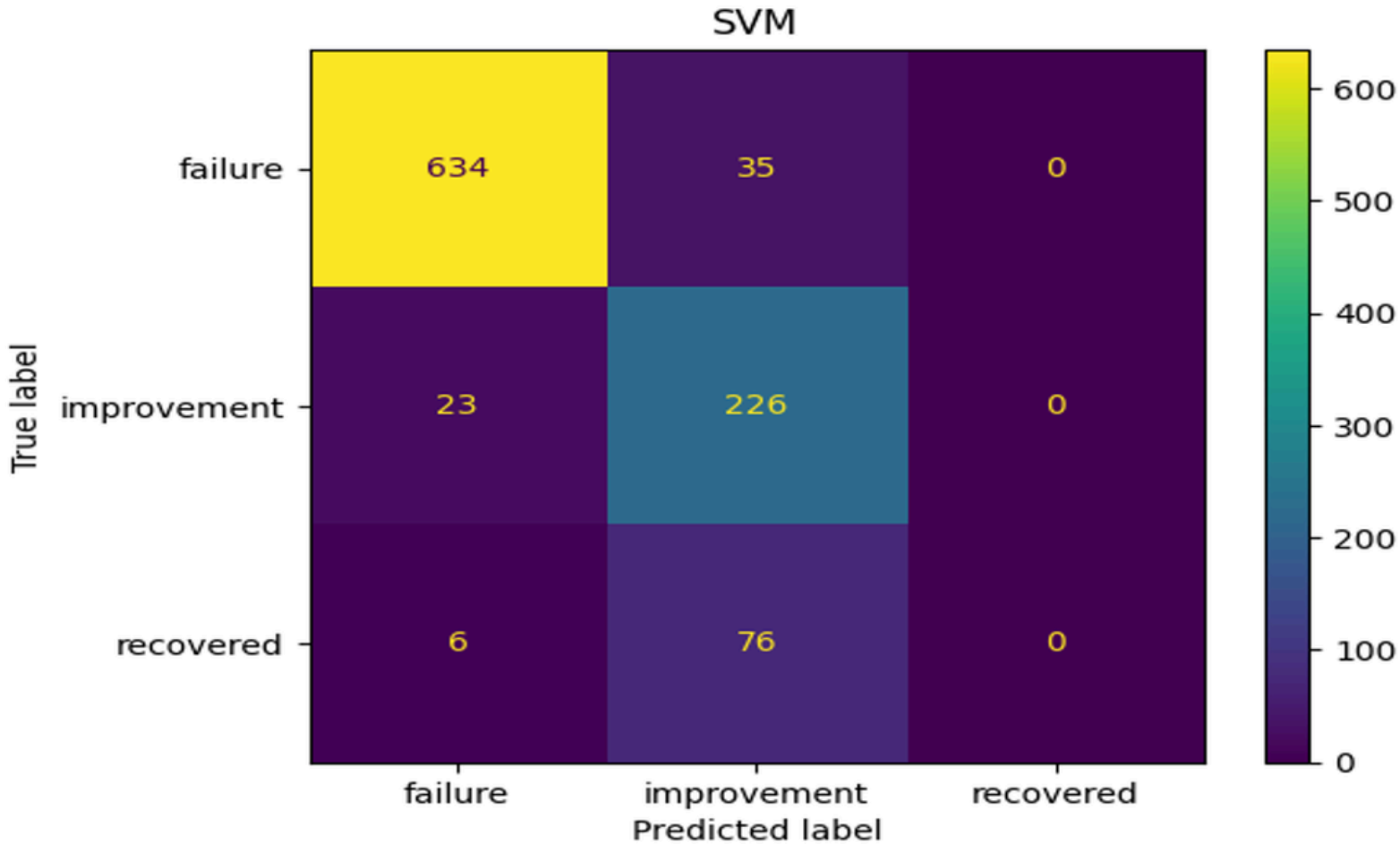
86.0%

ACCURACY

0.574

MACRO F1

SVM failed to predict the 'recovered' class at all (0% recall) — the minority class (8% of data) was completely missed. Feature scaling with StandardScaler was applied, but the model still struggled with class imbalance.



MODEL 5:RANDOMFOREST

SIMILAR TO XGBOOST BUT SIMPLER: IT BUILDS MANY DECISION TREES (100 HERE),
EACH TRAINED ON A RANDOM SUBSET OF DATA,
THEN THEY ALL VOTE ON THE FINAL ANSWER.
STRONG AND STABLE. RESULT: 99.8% — BEST SINGLE MODEL.

Random Forest

Multi-class classification of treatment outcome

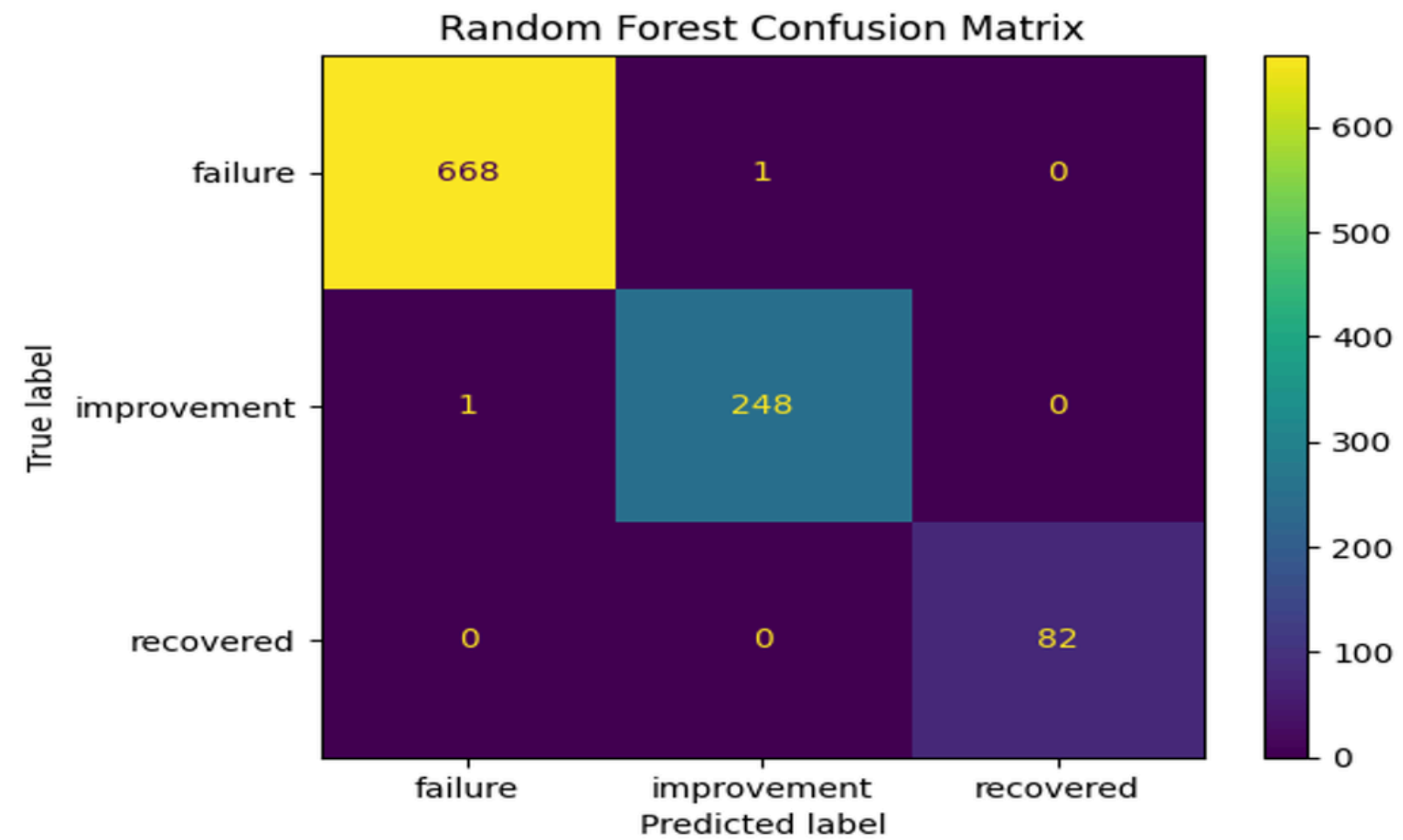
99.8%

ACCURACY

0.998

MACRO F1

Near-perfect performance across all three classes (failure, improvement, recovered), including the minority 'recovered' class. Strongest individual model in this comparison.



MODEL6:VOTING

Not a new model from scratch

- it combines 3 models (Logistic Regression + SVM + Random Forest) and lets them "vote" together using probabilities (soft voting) to make a more reliable final decision. Result: 97.1%.

Voting Classifier (Ensemble)

Soft voting: Logistic Regression + SVM + Random Forest

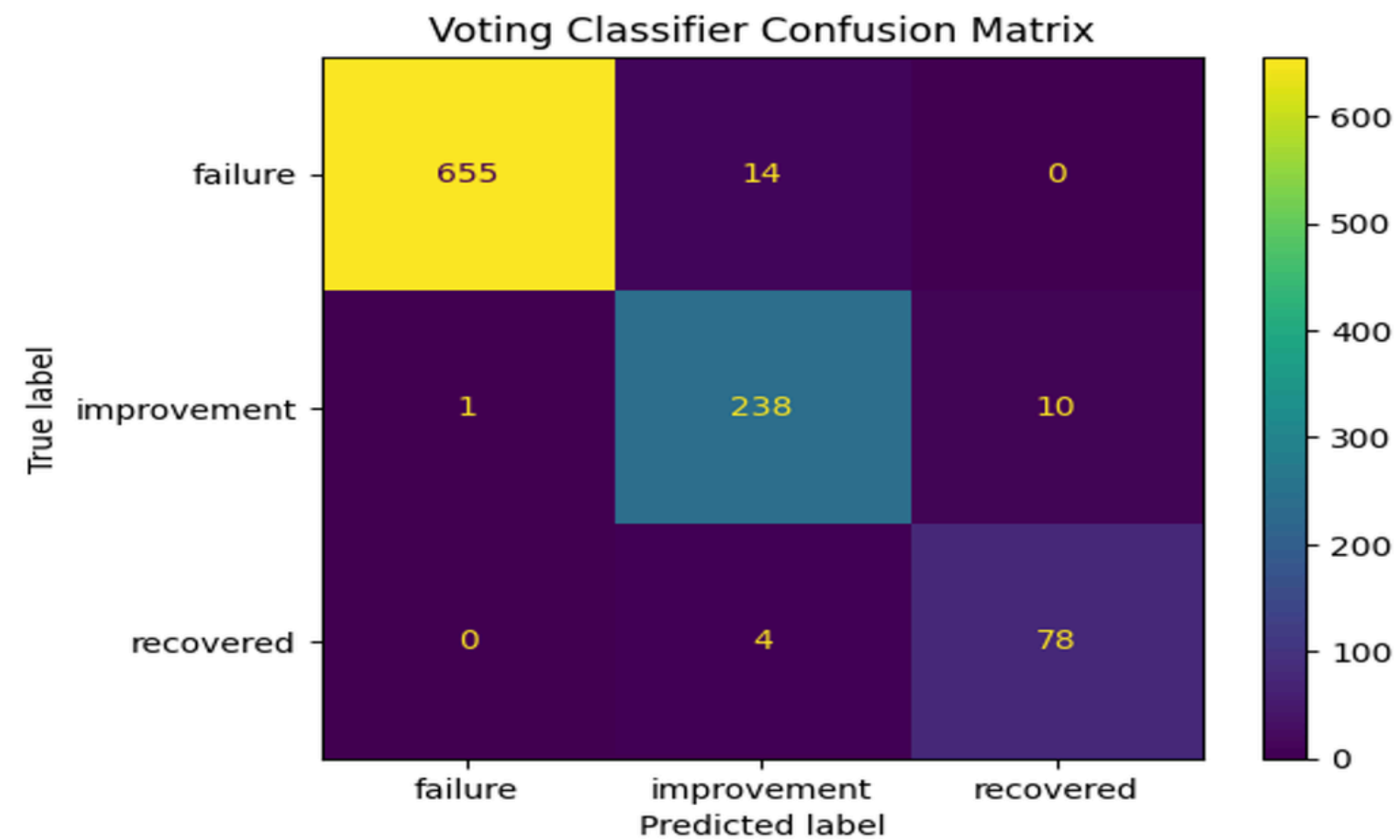
97.1%

ACCURACY

0.950

MACRO F1

Combines 3 models via soft voting (probability-weighted). Balanced performance across all classes, though slightly below Random Forest alone — SVM's weak recall on 'recovered' pulled the ensemble down a bit.



What the GUI Does

Four screens covering the full pipeline from raw data to a live prediction



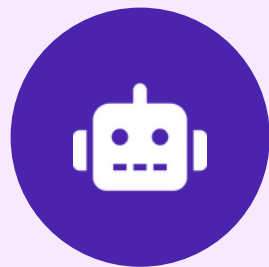
Upload & Overview

Upload the patient dataset and instantly see row/column counts, missing values, and the three categories in the data.



Exploratory Data Analysis

Reproduces every analysis and plot from the notebook — distributions, correlation heatmap, and outcome breakdowns.



Model Training & Comparison

Trains the same 6 models from the notebook (Logistic Regression, KNN, XGBoost, SVM, Random Forest, Voting Classifier) and compares their scores side by side.



NEW CAPABILITY

Predict a Case

The notebook only evaluates the model on its own test set. This screen goes further: it lets us predict an outcome for a brand-new patient who isn't in the dataset at all.

1

Enter patient data

Fill in a form with every field the models were trained on — vitals, labs, lifestyle, and history.

2

Choose a model

Pick any of the 6 trained models, or the top-performing Voting Classifier.

3

Get the outcome

The predicted `treatment_outcome` appears instantly, using the exact same encoding pipeline as training.



THANK YOU

Presented by Team 4

